24 March 1978

MILITARY SPECIFICATION

GENERATORS, SIGNAL

SG-1145( )/U

This specification is approved for use by the US Army Communications Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines both solid-state and vacuum tube type construction, super high frequency, signal generators which furnish low power radio frequency test signals at accurately adjustable frequencies and power levels. The continuous wave (cw) output can be frequency or pulse modulated. The generators are hereinafter called the equipment.

1.2 Classification. Equipment defined by this specification shall be of the following configurations:

1.2.1 Configuration A. Configuration A equipment shall be as specified in MIL-T-28800 for Type II, Class 3, Style B, Color R, and as herein.

1.2.2 Configuration B. Configuration B equipment shall be as specified in MIL-T-28800 for Type II, Class 5, Style E, Color R, and as herein with the convertible/rackmountable capability.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications Research and Development Command, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6625

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids, or the request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment General Specification for

STANDARDS

MILITARY

MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipments
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests, Exponential Distribution

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the equipment are classified herein as follows:

Requirement	Paragraph
Safety	3.2
Parts, Materials and Processes	3.3
Design and Construction	3.4
Electrical Power Sources and Connections	3.5
Dimensions, Weight	3.6
Enclosure Requirements	3.7
Marking & Identification	3.8

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Requirement	Paragraph
Environmental Requirements	3.9
Reliability	3.10
Performance Requirements	3.11
Inspection Requirements	3.12

3.2 Safety. Unless otherwise specified herein, the equipment shall comply with the Type II safety requirements specified in MIL-T-28800.

3.3 Parts, materials, and processes. Unless otherwise specified herein, the equipment shall comply with the Type II parts, materials and processes requirements of MIL-T-28800.

3.3.1 Restricted materials. Equipment shall comply with the restricted material requirements of MIL-T-28800 including the requirement for mercury or radioactive materials and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

3.4 Design and construction. Unless otherwise specified herein, the equipment shall comply with the Type II design and construction requirements of MIL-T-28800, and with the configuration requirements of this specification.
3.4.1 Mainframe plug-in concept. Equipment utilizing externally accessible, externally removable subassemblies (mainframe plug-in concept) do not meet the requirements of this specification and therefore are unacceptable. Items using this concept may be offered, provided the plug-in(s) shall not be removable by access through the front panel or rear panel with the unaided hand. A system part number shall be assigned to any such mainframe plug-in combination required to meet the requirements of this specification. The system part number shall be marked on the mainframe identification plate and any plug-in(s).

3.4.2 Controls. Unless otherwise specified herein, built-in adjustments and compensating devices shall not be externally accessible.

3.4.2.1 Front panel controls. All controls which are required to operate the equipment throughout its specified performance characteristics, shall be located on the front panel.

3.4.3 Accessibility. The equipment shall be constructed as follows:

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3.4.3.1 Major subassemblies. Major subassemblies and chassis components can be removed without removing any other hard wired subassembly, printed circuit card or component.

3.4.3.2 Adjustments. Adjustments can be made without removing any component, printed circuit card or subassembly except the use of extender cards is permitted.

3.4.3.3 Printed circuit cards. Printed circuit cards can be removed without the need to unsolder cables and interconnecting wiring (connections to all printed circuit cards shall be through pin and socket connectors). Printed circuit cards (mother boards) designed primarily to distribut power and signals to other printed cards (daughter boards) are excluded from this requirement. When such motherboards are used, they shall be accessible from both sides to allow maintenance testing.

3.4.3.4 Indicator lights. Unless approved by the procuring activity upon presentation of acceptable reliability data, indicator lights other than light emitting diodes (LEDs) shall be accessible from the operator's side of the front panel.

3.4.3.5 Encapsulation and embedment. Encapsulation and embedment (potting) of subassemblies shall not be used except for high voltage circuits.

3.4.4 Solid-state construction. Unless otherwise specified herein, the equipment shall be of solid-state, modular, miniaturized construction. (Vacuum tube type construction may be used as an alternate, if required).

3.4.5 Signal input and output connectors. Unless otherwise specified herein, all signal input and output connectors shall be Series "BNC" female, except for the RF output(s) which shall be a Series "N" connector, female.

3.5 Electrical power sources and connections. Unless otherwise specified herein, the equipment shall comply with the Type II electrical power sources and connections requirements of MIL-T-28800. The equipment shall operate from a nominal 115/230 volts single phase 50, 60, and 400 Hz source.

3.5.1 Maximum power. The maximum power consumption of the equipment shall be 250 watts.

3.5.2 Input power selection device. An input power selection device shall be provided for selection of input power voltages of 115 VAC or 230 VAC. Provision shall be incorporated to prevent accidental switching. When the equipment is delivered, the power selection device shall be in the 115 VAC position.

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3.5.3 Fuses and circuit breakers. Fuses or circuit breakers shall be in accordance with MIL-T-28800. (115 VAC/230 VAC) Either common or separate fuseholders may be provided. If only one fuseholder is used (common), the equipment shall be provided with the 115 VAC fuse installed and the 230 VAC fuse shall be stowed with the accessories.

3.5.4 Input power switch. A front panel mounted power switch shall be provided. The ON position shall have panel identification lights for AC operation. The switch shall break both sides of the power source.

3.5.5 External operating power connector. Operating power from an external source shall be furnished through a front panel connector for Style E enclosures. (A rear panel connector may be used as an alternate, if necessary.)

3.6 Dimensions and weight.

3.6.1 Dimensions.

3.6.1.1 Configuration A. The overall dimensions shall be 381mm (15 in.) maximum height, 517mm (20-3/8 in.) maximum depth, and 495mm (19 in.) maximum width.

3.6.1.2 Configuration B. The overall dimensions shall be 517mm (20-3/8 in.) maximum depth. The equipment width and height shall be as specified for Type II rack-mounted units in MIL-T-28800. A flat plate may be required to allow the converted unit to be used in a standard rack.

3.6.2 Weight. The weight for both Configurations (A and B) shall not exceed 28 kg (61.6 lbs).

3.7 Enclosure requirements. Unless otherwise specified herein, the equipment shall comply with the Style B or E enclosure requirements of MIL-T-28800 and the Configuration A and B requirements of this specification (See 6.2).

3.7.1 Stacking provisions. The maximum stacking height for Style B enclosures shall be 1.52 meters (60 inches).

3.8 Marking and identification. Unless otherwise specified herein, the equipment shall comply with the Type II marking and identification requirements of MIL-T-28800.

3.8.1 Supplemental identification plate. The supplemental identification plate specified in MIL-T-28800 shall contain the following data only:

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a. Nomenclature

b. Procurement instrument identification number (PIIN).

c. Serial number.

d. National stock number.

e. US.

3.9 Environmental requirements. Unless otherwise specified herein, the equipment shall comply with the Type II environmental requirements of MIL-T-28800 as applicable to its class.

3.9.1 Electromagnetic Interference (EMI). The signal generator shall comply with the following emission and susceptibility requirements of Notice 4, MIL-STD-461.

CE02	CS02	RE02.1	RS03
CE04	CS06	RE02	

RE02.1 and RS03 shall be performed from 14KHz to 10GHz with RS03 at a susceptibility level of one volt per meter (1 V/m).

3.9.2 Humidity. The equipment shall meet the humidity requirements in accordance with the test specified in paragraph 4.5.5.1.1.3 of MIL-T-28800B.

3.9.3 Vibration. The equipment shall comply with the Class 5, Style E vibration requirements of MIL-T-28800, except that the equipment need not be operating during vibration.

3.10 Reliability requirements.

3.10.1 Reliability burn-in. Each equipment delivered against this specification shall be submitted to a minimum 96 hour on-time burn-in procedure as specified in 4.4.5. The last 24 hours of burn-in shall be failure free.

3.10.2 Reliability. Reliability shall comply with requirements as specified herein. The specified MTBF shall be 2,500 hours when tested as specified in 4.4.4. A failure shall be defined in MIL-STD-781, para 3.1, and as any departure from the

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required performance or operation of the required accuracies (not correctable by normal use of the operating controls) after the test is initiated. Test Level B, MIL-STD-781 shall be the required test level.

3.10.3 Maintainability requirements. The equipment shall comply with the Type II maintainability requirements of MIL-T-28800.

3.11 Performance characteristics.

3.11.1 Output frequency.

3.11.1.1 Range. The output frequency range of the equipment shall, at a minimum, be 7 to 11 GHz.

3.11.1.2 Accuracy. The output frequency accuracy shall be within ±1% of the indicated frequency. This accuracy shall not depend upon individually calibrated indicators, calibration charts, curves, or tables.

3.11.1.3 Fine adjustment. The equipment shall have a frequency vernier control which will allow a fine adjustment of the output frequency of ±1.5MHz over most of the band.

3.11.1.4 Bands. The minimum required frequency range may be divided into a maximum of 6 bands.

3.11.1.5 Response. The output power level shall not vary more than 2 db over any indicated frequency band or when the output is switched from pulse modulated to unmodulated cw.

3.11.1.6 Voltage standing wave ratio (VSWR). The VSWR of the RF impedance of the equipment shall be less than 1.5:1.

3.11.1.7 Form. The RF output signals shall be available in the form of continuous-wave (cw), sawtooth frequency modulated (FM), pulse modulated (PM), and square wave modulated outputs.

3.11.1.8 Stability. The variation of continuous wave (cw) RF test signals shall not exceed the following under the noted conditions:

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3.11.1.8.1 Load variation. The output frequency shall not vary more than 0.001% as the load is changed from the rated value to any other value from open circuit to short circuit.

3.11.1.8.2 Temperature variation. The output frequency shall not vary more than 0.006% per degree Centigrade as the temperature is varied by any amount over the operating temperature range of the equipment.

3.11.1.8.3 Input power variation. The output frequency shall not vary more than 0.02% as the input power is changed from the nominal 115/230 volts, 60 Hz/400 Hz to any other combination of voltage and frequency within 10% of the nominal input values.

3.11.2 Output power. The output power level of the cw RF test signals shall be continuously variable from -127 dbm to 0 dbm.

3.11.2.1 Output power indication error. The error of the direct indication of the output power shall be less than 2 dbm of the indicated value for output levels from -7 dbm to -127 dbm and less than 3 dbm for output levels from 0 to -7 dbm when applied to a load equal to rated output impedance.

3.11.3 Output impedance. The output impedance of the equipment shall be 50 ± 2 ohms.

3.11.4 Internal pulse modulation. The modulation circuitry shall be capable of modulating the RF oscillator using pulses having the following characteristics:

3.11.4.1 Pulse rate. The pulse rate shall be continuously adjustable within 40 to 4000 pulses per second.

3.11.4.2 Pulse width. The pulse width shall be continuously variable within 0.5 microseconds to 10 microseconds.

3.11.4.3 Pulse rise/fall time. The pulse rise time and the pulse fall time shall each individually be less than 0.2 microseconds.

3.11.5 External pulse modulation. Provision shall be included to allow the RF test signals to be pulse modulated from an external source with pulses having the following characteristics:

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3.11.5.1 Pulse amplitude. External pulses shall have an amplitude between 20 and 70 volts, positive or negative polarity.

3.11.5.2 Pulse width. The pulse width shall be variable from 0.5 to 2500 microseconds with pulse separation from 1.5 to 2500 microseconds.

3.11.5.3 Pulse rise/fall time. The pulse rise time and the pulse fall time shall each individually be less than 0.2 microseconds.

3.11.6 Modulation power. The peak output power of the pulse modulated RF test signals shall be equal to the average unmodulated cw power within ±1.0db.

3.11.7 Internal frequency modulation. The internal sawtooth frequency modulated (FM) test signals shall have the following characteristics:

3.11.7.1 Rate. The FM rate shall be adjustable in the range of 40 Hz to 4000 Hz.

3.11.7.2 Deviation. The FM deviation shall be adjustable within the range of 0 - 5 MHz over most of the frequency band.

3.11.8 Square wave modulation. The equipment shall permit the RF test signals to be square wave modulated by an internal source whose frequency is continuously variable from 40 Hz to 4000 Hz.

3.11.9 Output attenuation. An output attenuator shall be provided with its control calibrated to indicate the output RF power level. The control shall be calibrated in decibels (-127 db to 0 db) and in voltage (0.1 microvolt to 233 millivolts).

3.11.10 Distortion. Distortion of the output signals shall be limited to the degree shown following for the specific types noted:

3.11.10.1 Spurious signals. The total power of all sub and non-harmonically related spurious signals within the range of 1,000 to 22,000 MHz accompanying the desired cw RF test signal shall be at least 60 db below the power of the desired RF signal, excluding line frequency related sidebands on the desired RF test signal.

3.11.10.2 Harmonics. The power of any harmonic of the desired continuous-wave RF test signal shall be at least 30 db below the power of the desired RF test signal.

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3.11.10.3 Residual FM. The residual, uncontrolled frequency modulation of the intentionally unmodulated cw RF test signals shall not cause a deviation of frequency in excess of 15 KHz peak.

3.11.10.4 Residual AM. The residual uncontrolled amplitude modulation of the intentionally unmodulated cw RF test signals shall not exceed 0.5 percent.

3.11.11 Internal synchronization. The equipment shall be capable of providing synchronization signals to the modulated output test signals. The synchronizing signals shall have the following characteristics:

3.11.11.1 Amplitude. The synchronizing pulses shall be positive only and shall be variable from 25 to 100 volts.

3.11.11.2 Width. The synchronizing pulses shall have a continuously variable pulse width from 0.5 to 5 microseconds.

3.11.12 External synchronization. It shall be possible to synchronize both pulse and frequency modulated test signals with external signals with the following characteristics:

3.11.12.1 Sine waves. The equipment shall be capable of being synchronized by an external sine wave signal whose frequency range is 40 Hz to 4000 Hz and whose amplitude is 5 to 50 volts rms.

3.11.12.2 Pulse signal. The equipment's RF output signal shall be capable of being synchronized by an external pulse signal which has a repetition rate from at least 40 pps to 4000 pps, a pulse width within the range of .5 to 5μsec and a peak amplitude within the range of 5 to 50 volts.

3.11.13 Synchronization delay. Synchronization pulses, internal and external, shall be capable of being delayed in a continuously adjustable manner in the range of 3 to 300 microseconds.

3.11.14 Output signals. Output synchronizing and modulating signals (sawtooth, delay, undelayed, FM, and pulse) shall be available for use in external equipment.

3.12 Inspection requirements.

3.12.1 First article inspection. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with para 4.3 (see para 6.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections required herein are classified as in a and b.

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.3).

4.3 First article, or quality conformance inspection. Unless otherwise specified herein, the first article and quality conformance inspections shall be in accordance with MIL-T-28800.

4.3.1 Test Plan. Unless otherwise stated in the contract, the contractor shall prepare a test plan in accordance with paragraphs 4.3(a) and 4.4(a) of MIL-T-28800B, for use with both classes of inspection. The test plan shall include as a minimum the tests listed in TABLES I, II, & III, the tests noted in section 4.4, and a description of the satisfactory operation check and satisfactory operation test, as defined in para. 4.5 of MIL-T-28800B. Unless otherwise required, the tests and inspections to be performed shall be identical for both classes of inspection.

4.3.2 Inspection sampling plan. The inspection sampling to be performed during both classes of inspection shall be as specified in MIL-T-28800, except as indicated below:

4.3.2.1 Unless otherwise specified herein or in the contract, the Group D, C, E & F quality conformance inspections shall be performed on the first production lot only.

4.3.2.2 The Group D EMI tests and Group F Reliability test shall be performed during quality conformance inspection only if not performed during first article or bid sampling testing.

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TABLE I

EXAMINATION AND TEST GROUPS

Description	Requirement	Test Method
Group A Preoperational Inspection Leakage current Level A performance	4.5.3.1 * 3.2 TABLE II	4.3.1
Group B Level B performance	TABLE III	4.3.1
Group C Electrical power Environmental requirements Humidity Vibration	3.5 3.9 3.9.2 3.9.3	4.3.1
Group D Electromagnetic interference	3.9.1	4.4.1
Group E Dimensions Weight Front panel marking	3.6.1 3.6.2 3.8	4.3.1
Group F Reliability	3.10	4.3.1, 4.4.2

* Paragraph referenced is in MIL-T-28800

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TABLE II

LEVEL A PERFORMANCE TESTS 1/

Description	Requirements
Power input	3.5
Power switches, input and selection	3.5.2, 3.5.4
Output frequency range	3.11.1.1
Output power range	3.11.2
Output impedance	3.11.3
Internal pulse modulation	3.11.4
External pulse modulation	3.11.5
Internal frequency modulation	3.11.7
Internal square wave modulation	3.11.8
Output attenuation	3.11.9
Internal synchronization	3.11.11
External synchronization, sine wave	3.11.12.1
External synchronization, pulse	3.11.12.2

1/ Level A testing is abbreviated testing (see 6.4.1)

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TABLE III

LEVEL B PERFORMANCE TESTS

Description	Requirements
Output frequency range	3.11.1.1
Output frequency accuracy	3.11.1.2
Output frequency fine adjustment	3.11.1.3
Output frequency response	3.11.1.5
Standing save ratio (SWR)	3.11.1.6
Electrical stability	3.11.1.8
Output power range	3.11.2
Output power indication error	3.11.2.1
Output impedance	3.11.3
Internal pulse modulation rate	3.11.4.1
Internal pulse modulation width	3.11.4.2
Internal pulse modulation rise/fall time	3.11.4.3
External pulse modulation amplitude	3.11.5.1
External pulse modulation width	3.11.5.2
External pulse modulation rise/fall time	3.11.5.3
Internal frequency modulation	3.11.7
Square wave modulation	3.11.8
Distortion, spurious signals	3.11.10.1
Distortion, harmonics	3.11.10.2
Distortion, incidental FM	3.11.10.3
Distortion, incidental AM	3.11.10.4
Internal synchronization, amplitude	3.11.11.1
Internal synchronization, width	3.11.11.2
External synchronization, sine wave	3.11.12.1
External synchronization, pulse	3.11.12.2

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4.4 Quality assurance tests. Unless otherwise specified herein, the following tests shall be performed in accordance with MIL-T-28800.

4.4.1 Electromagnetic interference. One equipment shall be subjected to an EMI test in accordance with MIL-STD-461 and the requirements of 3.9.1. All test set-ups and procedures shall comply with the measurement techniques of Notice 3, MIL-STD-462. All emission and susceptability tests shall be performed with the output cable connected to, and extended parallel with, the front of the equipment. The output cable shall be terminated to simulate a normal loading configuration.

4.4.2 Humidity. The equipment shall be subjected to the humidity test specified in paragraph 4.5.5.1.1.3 of MIL-T-28800.

4.4.3 Vibration. The equipment shall be subjected to the vibration tests specified in MIL-T-28800, for Class 5 equipment, except that the equipment shall not be operated during the test. The satisfactory operation test shall be performed prior to and following vibration testing.

4.4.4 Reliability test. The Group F reliability tests shall be performed as follows:

4.4.4.1 Reliability sampling plan.

4.4.4.1.1 First Article. The Group F reliability demonstration shall be conducted on at least five (5) samples. The use of ten (10) samples is encouraged, but not more than ten (10) samples shall be used.

4.4.4.1.2 Quality conformance inspection. The Group F reliability tests shall be performed on the first production lot, only if not performed during First Article testing. From those units of the first lot that have passed the required Group A and B tests, a random sample of ten (10) units shall be selected, for Group F testing.

4.4.4.2 Reliability test plan. The reliability tests shall be conducted in accordance with the following test plan:

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θ_o = 2,500 hours

Number of Failures	Total Test Time * Accept ** (Equal or more)	Reject *** (Equal or less)
0	2,000	N/A
1	3,750	558
2	5,375	2063
3	6,875	3850
4	8,375	5725
5	9,875	7725
6	11,375	11375 ****

* Total test time is total unit hours of "equipment-on" time.
** Accept if test time is greater than or equal to that listed.
*** Reject if test time is less than or equal to that listed.
**** Reject if test time is strictly less than 11,375 with 6 failures.

4.4.4.3 Test length. Testing will continue until the total unit hours together with the total count of relevant equipment failures permit either an accept or reject decision in accordance with the specified test plan. Only equipment "ON" time may be used in MTBF or longevity determinations. Testing shall be monitored in such a manner that the times to failure may be estimated with reasonable accuracy. No single equipment "ON" time shall be less than one-half the average operating time of all equipments on test.

4.4.4.4 Test conditions.

4.4.4.4.1 Operating mode (Duty Cycle). Prior to the burn-in, a satisfactory operation test shall be conducted. Daily satisfactory operation checks shall be conducted. For the last 24 hour failure free period, a complete satisfactory operation test shall be conducted prior to and after the period.

4.4.4.4.2 Test level. While under test, the equipment shall be subjected to the conditions specified by Test Level B, MIL-STD-781.

4.4.4.4.3 Input voltage cycling. When so directed by the procuring activity, voltage cycling shall be accomplished as follows: The input voltage shall be maintained

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at one hundred ten percent (110%) nominal voltage for one-third of the equipment "ON" cycle, at the nominal value for the second one-third of the equipment "ON cycle, and at ninety percent (90%) for the final one-third of the equipment "ON" cycle. This cycling procedure is to be repeated continuously throughout the reliability test.

4.4.4.5 Reliability examination and test method. The following inspections shall be used to verify equipment operation during reliability testing. The satisfactory operation test of 4.3.1 shall be used to confirm proper equipment operation prior to and following reliability testing. The satisfactory operation test shall also be performed weekly to verify equipment operation. The satisfactory operation check of 4.3.1 shall be used to monitor proper operation of the equipment daily.

4.4.4.6 Reliability failure actions, Group F. In the event of failure(s) during reliability testing, failure action shall be taken as required in MIL-STD-781, para 5.5.

4.4.4.7 Corrective action, Group F. In the event the reliability test reaches a reject decision, corrective action shall be taken as required in MIL-STD-781, para 5.7.

4.4.5 Burn-in. Each deliverable equipment shall be subjected to a minimum 96 hours on-time burn-in period, prior to Group A testing. During the last 24 hours of burn-in, the equipment must operate failure free. Up until this time, equipment will be allowed to accumulate failures. Each equipment which fails during the final 24 hour period shall be repaired and returned to test until it successfully survives a 24 hour period without failure. Failures which occur during the burn-in test shall be noted and reported but shall not count toward the establishment of equipment MTBF.

4.4.5.1 Burn-in conditions. Test Level B of MIL-STD-781 shall constitute the required test conditions. The input voltage cycling procedure per para 5.2.4 shall be invoked.

4.4.5.2 Burn-in tests. Prior to burn-in, the satisfactory operation of 4.3.1 shall be conducted. Daily satisfactory operation checks (4.3.1) shall be conducted. For the last 24 hour failure free period, a complete satisfactory operation test shall be conducted prior to and after the period.

4.4.6 Optional quality assurance system. The procuring activity may substitute a Department of Defense (DoD) approved supplier's quality assurance system for any or all tests in TABLE I.

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5. PACKAGING

5.1 Preservation, packaging. Packaging for delivery shall be in accordance with MIL-T-28800 and as specified by the procuring activity.

6. NOTES

6.1 Intended use. The equipment is intended for use in maintaining electronic communications equipment by equipment repair units in intermediate and depot maintenance.

6.1.1 Equipment replaced. Equipment procured accordance to this specification is intended to replace all versions of various older equipments now fielded. These equipments to be replaced are listed below:

AN/URM-44 AN/USM-190 SG-944

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.
b. Packaging requirements (see Section 5).
c. When rough handling and functional tests are required.
d. Place of final inspection.
e. Technical literature required.
f. Quantity of tools and running spare parts required.
g. Marking for shipment and shipping containers.
h. Test plans and test reports.
i. Classification of inspection and number of samples required.
j. Rack mounting requirements.
k. Alternate power requirements.
l. Configuration requirements.
m. Maintainability Rationale.
n. Reliability Rationale.

6.3 Contract data requirements:

a. Nomenclature assignment.
b. National stock number.
c. Equipment sample test plans.
d. Pretest performance records.
e. First production test data.
f. Identification plate drawing.

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6.4 Definitions.

6.4.1 Level A performance tests. Level A testing (TABLE II) is a reduced amount of testing which is performed on each equipment produced. Its purpose is to insure that all functions and modes of operation of the equipment are evaluated without extensively checking each parameter as required in the Level B test (TABLE III). The approved equipment test procedure shall specify the actual amount of testing to be performed.

6.4.2 Reliability rationale. Reliability rationale submitted with the sample should provide clear and concise rationale showing how the reliability of the equipment complies with minimum requirements of the solicitation. This may include data from previous reliability tests, reliability predictions and other data available to the offeror. In the absence of such data on the equipment being offered, such data on similar reuipment of equal or greater complexity produced by the offeror may be submitted. However, such data must be clearly identified as comparative data, and accomplished with specifications and technical literature on the similar equipment. The acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing the production reliability testing.

6.4.3 Maintainability rationale. Maintainability rationale submitted with the sample should provide clear and concise rationale showing how the maintainability of the equipment complies with requirements of the solicitation. This may include data from previous maintainability tests, records of repair and calibration data, and other data on the equipment available to the offeror. Data on selection of module size and other design features relevant to maintainability may also be submitted. The acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing the production maintainability testing.

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